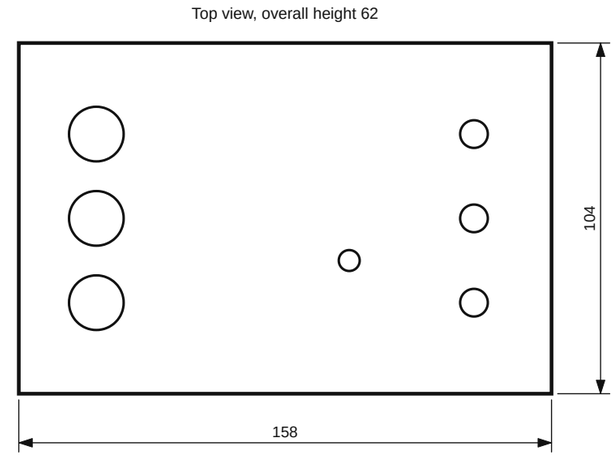
POWER GRID BOX

3D printed enclosure - layout sketches

Drawing PGB-ENC-001 | 2026-11-09 | all dimensions in mm



MAIN DIMENSIONS

External size (W x D x H)	**158 x 104 x 62 mm**
Internal size (W x D)	**153 x 99 mm**
Base height + lid	**59 + 3 mm**
Wall / floor / lid thickness	**2.5 / 2.5 / 3 mm**
Internal volume (approx.)	**0.86 L**
Top panel	**3 pushbuttons (left), selector SW1, 3 LEDs (right)**
Inside, left	**9 V battery holder, flat, long side left-right**
Inside, right	**Protoboard 30 x 33 above Arduino Uno R3**
Power	**9 V holder lead -> DC plug into Uno jack**

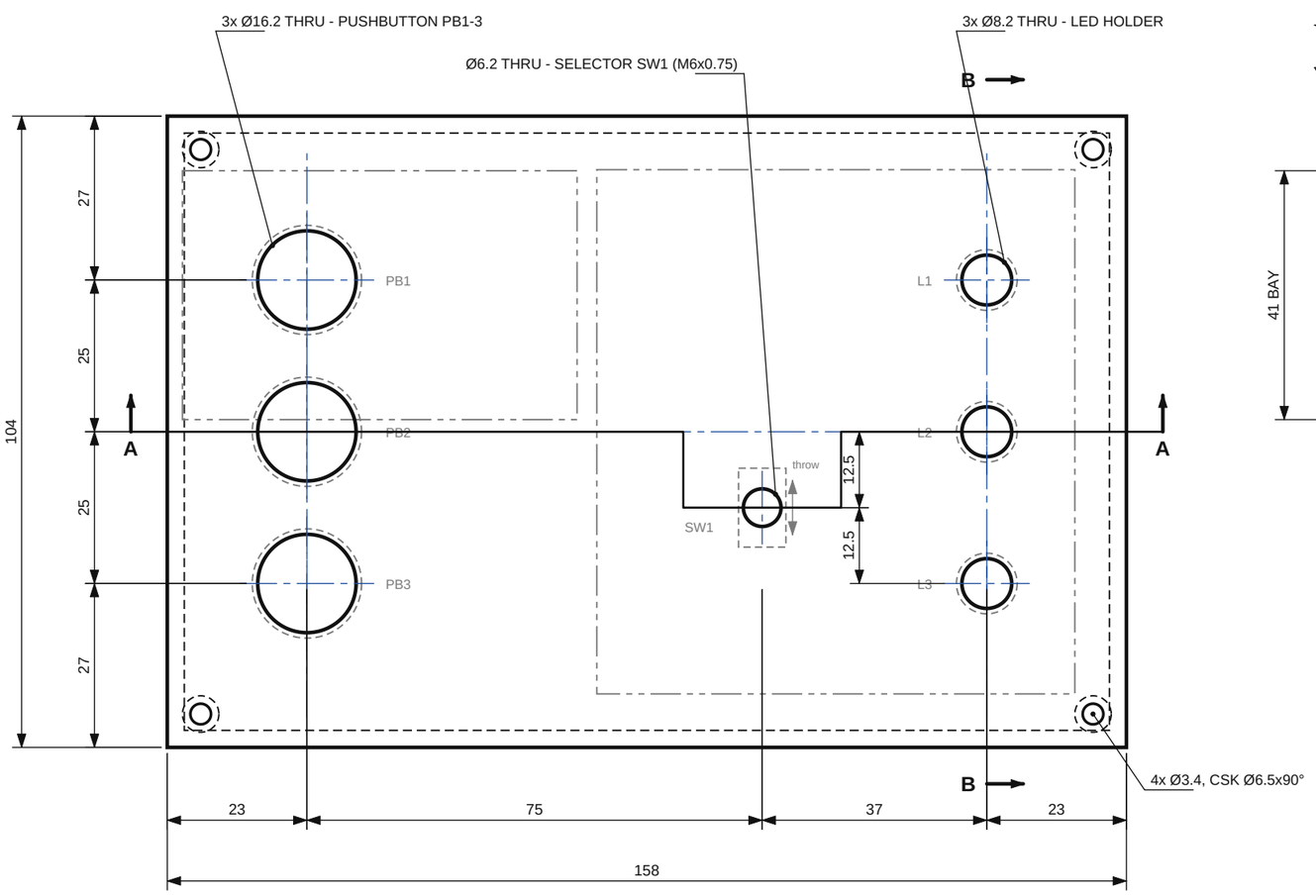
CONTENTS

Sheet 1	Cover - enclosure overview
Sheet 2	General arrangement - top panel, plan with lid removed, feature coordinates, legend
Sheet 3	Sections A-A / B-B and vertical stack-up

PROJECT: **POWER GRID BOX - 3D PRINTED ENCLOSURE**			DRAWING No.: **PGB-ENC-001**
TITLE: **COVER - ENCLOSURE OVERVIEW**			SHEET: **1 OF 3**
UNITS / SCALE: mm 1:1 @ A3	REV: D - PRELIM.	DATE: 2026-11-09	PROJECTION
MATERIAL / PROCESS: PETG, FDM, 0.2 mm layers	GENERAL TOL.: +/-0.3 unless stated	STATUS: **PRELIMINARY - FOR REVIEW**	

VIEW 1 - TOP (CONTROL PANEL / LID)

Lid 3.0 thick. Hidden: lid locating lip. Phantom: bays below.



VIEW 2 - PLAN, LID REMOVED

Holder long side left-right, back-left | electronics RIGHT under LEDs

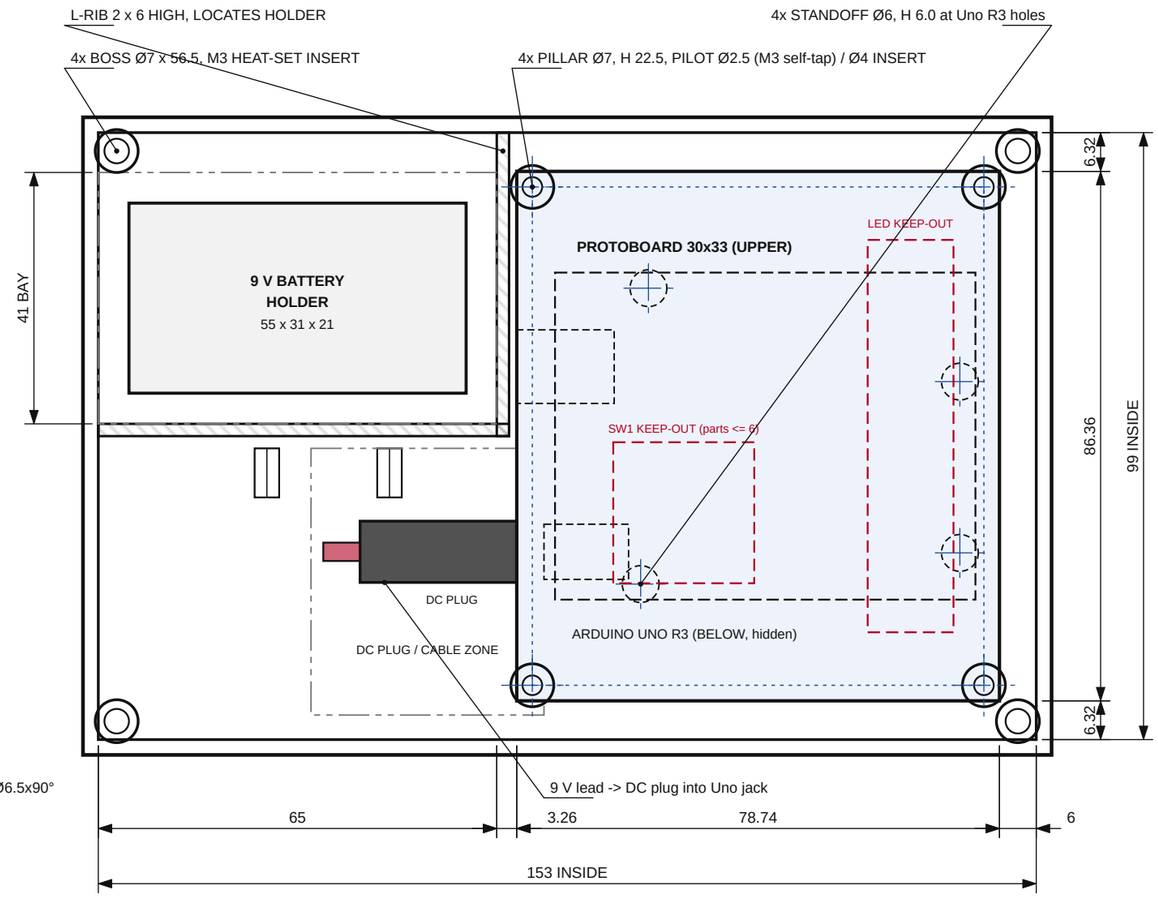


TABLE 1 - FEATURE COORDINATES (origin: external front-left-bottom corner, mm)

ITEM	X	Y	SIZE	NOTE
PB1/PB2/PB3	23	77 / 52 / 27	Ø16.2	Adafruit 1439, 16 mm illum. momentary
SW1 selector	98	39.5	Ø6.2	Miyama MS-500E (ON)-OFF-(ON), M6x0.75
L1/L2/L3	135	77 / 52 / 27	Ø8.2	LED + 8 mm panel holder (model TBC)
Lid screws	5.5 / 152.5	5.5 / 98.5	Ø3.4	4x M3 x 10 countersunk into inserts
Battery holder	7.5-62.5	59-90	55x31x21	flat, long side along X
Protoboard	70.76-149.5	8.82-95.18	78.74x86.36	underside Z 25
PB pillars	73.3 / 146.96	11.36 / 92.64	Ø7 x 22.5	corner holes drilled to Ø3.2
Arduino Uno R3	77-145.58	25.33-78.67	68.58x53.34	underside Z 8.5, USB/DC jack face LEFT
Uno standoffs	90.97 / 92.24 / 143.04 / 143.04	27.87 / 76.13 / 32.95 / 60.89	Ø6 x 6.0	4 holes, ref. Uno R3 drawing

LINE LEGEND

- Visible outline
- Hidden edge
- Centre line
- Zone / phantom (under lid)
- Keep-out zone
- Cut material (sections)
- Protoboard (upper level)

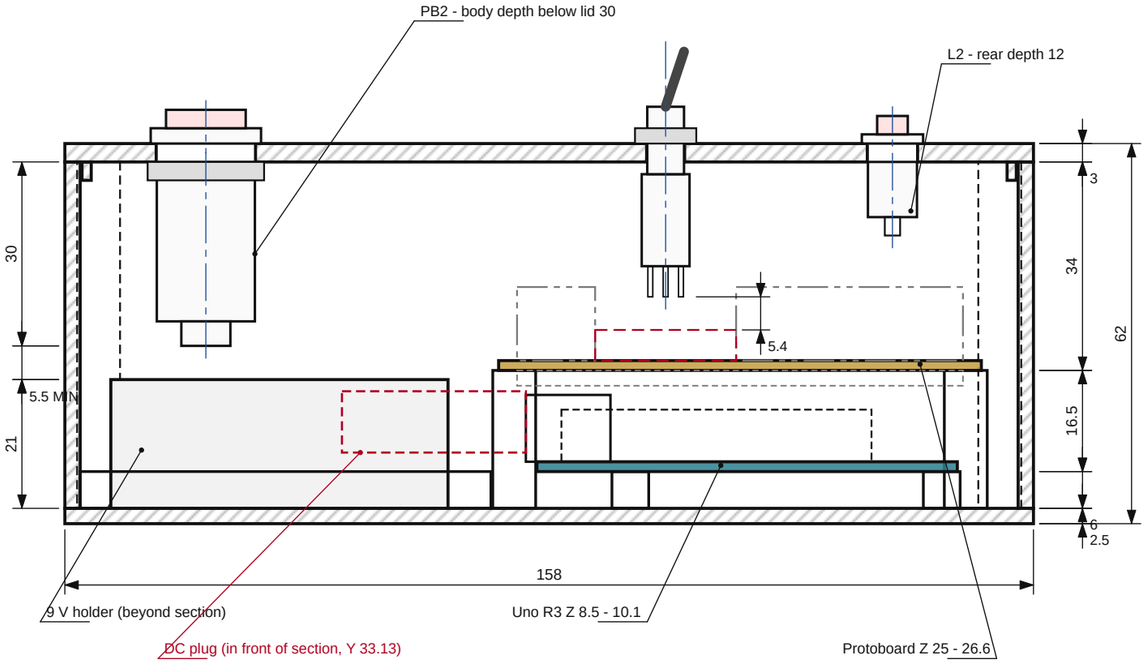
MAIN DIMENSIONS

External (W x D x H)	**158 x 104 x 62**
Internal (W x D)	**153 x 99**
Base height / lid	**59 / 3**
Wall / floor	**2.5 / 2.5**
Lid screws	**4x M3, corner bosses Ø7**

PROJECT: **POWER GRID BOX - 3D PRINTED ENCLOSURE**			DRAWING No.: **PGB-ENC-001**
TITLE: **GENERAL ARRANGEMENT - PLAN VIEWS**			SHEET: **2 OF 3**
UNITS / SCALE: mm 1:1 @ A3	REV: D - PRELIM.	DATE: 2026-11-09	PROJECTION
MATERIAL / PROCESS: PETG, FDM, 0.2 mm layers	GENERAL TOL.: +/-0.3 unless stated	STATUS: **PRELIMINARY - FOR REVIEW**	

SECTION A-A (stepped, viewed from front)

Stepped section: Y = 52 (PB2, L2), jog to Y = 39.5 through SW1



SECTION B-B (X = 135, viewed from right)

Through LED column L1-L3 and the board stack

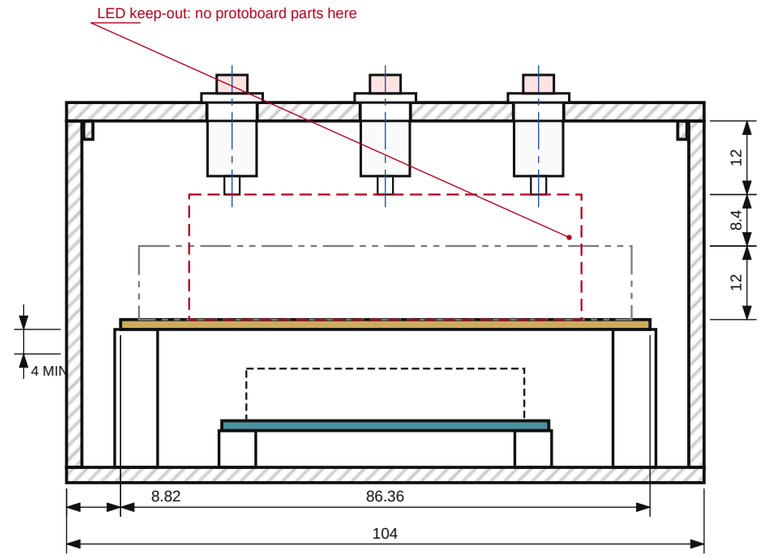


TABLE 2 - VERTICAL STACK-UP (Z from outside bottom)

LAYER	Z RANGE	NOTE
Floor	0 - 2.5	2.5 floor, 4x Ø10 x 1 foot recesses
Uno standoffs	2.5 - 8.5	6.0 high, Ø6
Uno R3 PCB	8.5 - 10.1	1.6 PCB
Uno parts / headers	10.1 - 21.0	DC jack/USB 10.9, headers 8.5
Clearance	21.0 - 25	4 to Uno parts, 3.4 to DC plug
Protoboard	25 - 26.6	on 4 floor pillars, H 22.5
Protoboard parts	26.6 - 38.6	max 12 (keep-out under LEDs)
LED rear	47 - 59	12; 8.4 clear to parts
Battery holder	2.5 - 23.5	21 high, on floor
Selector rear	37 - 59	22; parts under SW1 <= 5
Button rear	29.0 - 59.0	30; 5.5 clear to holder
Base top / lid	59.0 - 62.0	3.0 lid, 1.5x3 lip, 0.3 gap

LEGEND

- Visible outline
- Hidden edge
- Centre line
- Zone / phantom (under lid)
- Keep-out zone
- Cut material (wall, floor, lid)
- 9 V battery holder
- Protoboard
- Arduino Uno R3

PROJECT: POWER GRID BOX - 3D PRINTED ENCLOSURE			DRAWING No.: PGB-ENC-001
TITLE: SECTIONS A-A / B-B, VERTICAL STACK-UP			SHEET: 3 OF 3
UNITS / SCALE: mm 1:1 @ A3	REV: D - PRELIM.	DATE: 2026-11-09	PROJECTION
MATERIAL / PROCESS: PETG, FDM, 0.2 mm layers	GENERAL TOL.: +/-0.3 unless stated	STATUS: **PRELIMINARY - FOR REVIEW**	